

Information Systems

Data Management for the Web

IDENTIFICATION

CODE : IF-3-DW
ECTS : 2.0

HOURS

Lectures : 9.0 h
Seminars : 4.0 h
Laboratory : 8.0 h
Project : 0.0 h
Teacher-student
contact : 21.0 h
Personal work : 30.0 h
Total : 51.0 h

ASSESSMENT METHOD

Lab evaluation
Individual written exam

TEACHING AIDS

Lecture notes, Lab and exercise

TEACHING LANGUAGE

English

CONTACT

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AIMS

This course aims to address the problems related to XML technologies[DTD, Schema, XPath, XSL,XQuery, DOM, Javascript] and the NoSQL dbms

Skills :

- design and model XML documents
- create a DTD and an XML Schema
- write XPath expressions
- write complex XSL stylesheets
- write javascript functions to manipulate html DOM
- call REST webservices and manipulate XML and json results in javascript
- install and use XML DBMS (BaseX)
- install MongoDB
- write MongoDB queries

CONTENT

1. Context and issues: survey of major concepts and domains of application.
2. Documents and multimedia hyper documents: Modeling of specific documents, class of documents, hyper documents, and multimedia contents.
3. Introduction to XML: major concepts [element, attribute, entity], document type definition [DTD] and instance.
4. Advanced XML Concepts: name-space domains, schema, XLINK links, style sheet processing [XSL].
5. The XML galaxy: RDF, DOM, SVG.
6. Semi-structured databases: Representation of semi-structured data, XML and semi-structured data, XML databases.
7. XML manipulation: DOM, SAX, Javascript
8. Conclusion, assessment and perspectives.
9. NoSQL DBMS in general
10. Specific introduction on document based DBMS-s through MongoDB

PRE-REQUISITE

IF-3-BDR, IF-3-MD
Basic Relational Database Management notions